Rational Number Project

Initial Fraction Ideas Lesson 8: Overview	Materials
Students explore fraction equivalence by naming fractions equal to 1/2 with fraction circles and by finding other fraction equivalences with fraction circles.	• Fraction Circles for students and teacher • Student Pages A and B for students

Teaching Actions

Warm Up

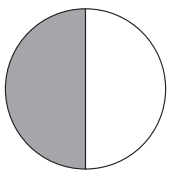
Use paper folding strips to show these two fractions: $\frac{3}{4}$ and $\frac{1}{2}$. Compare the strips – which fraction is larger? Why?

Large Group Introduction

1. Start the lesson with a quick review of naming fractions using the fraction circles. Set the stage for this by telling students that the whole circle is our unit, the amount to be divided into equal parts. Ask: If the whole circle is our unit, what fraction name is 1 blue? 2 browns? 6 grays? 4 pinks?

2. If students struggle naming the fraction amount ask: how many blues cover the whole circle? What fraction is one blue?

3. At the overhead, cover the whole circle with 1 yellow and ask students to find different ways to cover the remaining half of the circle. Record answers by color.

Ex: 

1 blue and 2 grays
2 blues
•
•
•
•
•

4. Repeat this activity, but specify that they have to use the same color pieces to cover half the circle. Record

Comments

The idea of equivalence is a prerequisite for fraction operations. To add $\frac{1}{2}+\frac{3}{4}$, you will explain that $\frac{1}{2}$ can be exchanged for $\frac{2}{4}$ because $\frac{1}{2}=\frac{2}{4}$.

Equality should first be developed from concrete models before explaining a rule that generates equal fractions $\left(\frac{1}{2}\times\frac{2}{2}=\frac{2}{4}\right)$

You are defining equivalent fractions by showing that fractions are equivalent if they cover the same amount of the circle. Partitioning is different so the digits in the fraction symbols are different; but 1 of 2 equal parts covers the same amount as 2 of 4 equal parts.

One-half equivalences are the most important ones for children to learn. Students with good "fraction number sense" use $\frac{1}{2}$ as a reference point for estimating the size of other fractions.

A student will use his/her concept of $\frac{1}{2}$ to estimate, for example,

Teaching Actions

results by color and fraction name.

1 yellow	=	2 blues	$\frac{1}{2}=\frac{2}{4}$
1 yellow	=	3 pinks	$\frac{1}{2}=\frac{3}{6}$

5. Ask what each display has in common. They all cover 1-half of the black circle therefore they are all **equivalent**. Introduce the term, equivalence.

Small Group/Partner work

6. Students continue to explore fraction equivalences for fractions other than ½. Student Page A is a review for naming fractions using fraction circles; Student Page B, asks students to concretely find fraction equivalences and to record them with words and then symbols.

Wrap Up

7. At this point it the lessons you should assess which students have strong mental images for fractions based on the fraction circles. Ask the following questions. Have students describe their mental representations:
    a. If the whole circle is our unit, what color represents thirds?
    b. What color represents sixths?
    c. Which is bigger one brown or one pink?
    d. Which is bigger 1/3 or 1/6?
    e. What color is fourths?
    f. Which is bigger ¼ or ¾?

8. Ask students to picture the fraction circles in their mind: Which fractions do you see that are equal to ½? What colors do you see? How many of each color covers ½ exactly?

Comments

$\frac{2}{4}+\frac{2}{3}$: " $\frac{2}{4}$ equals $\frac{1}{2}$; $\frac{2}{3}$ is greater than $\frac{1}{2}$; so the sum is greater than 1."

This reasoning is an excellent example of thinking quantitatively. Thinking quantitatively is not rule-bound, but relies on the mental images children have for fractions. Images for $\frac{1}{2}$ are critical.

Just as students learn the basic +, -, x, ÷ facts, you want them to learn basic fraction equivalents for $\frac{1}{2},\frac{1}{3},\frac{2}{3},\frac{1}{4},\frac{2}{4},\frac{3}{4}$.

On this level, they don't need rules, but many physical/visual examples with these fractions and their equivalents.

Translations

- Written symbols to manipulative to written symbols
- Manipulatives to verbal

Use paper folding strips to show these two fractions:

$$\frac{3}{4} \text{ and } \frac{1}{2}$$

Compare the strips – which fraction is larger? Why?

Grade

Naming Fractions

Directions: Use the **black circle** as the unit. Name the fraction amount for the number of pieces shown below.

1 yellow = $\frac{1}{2}$	2 browns =
1 blue =	2 oranges =
2 reds =	2 yellos =
6 grays =	0 blues =
4 pinks =	3 yellows =

Practicing Fraction Equivalence with Our Fraction Circles

Show the Equal Fractions with your Fraction Circles	Record the Equivalent Fractions with Symbols
1 yellow = _____blues	$\frac{1}{2} = _$
1 yellow = _____pinks	$\frac{1}{2} = _$
1 yellow = _____grays	$\frac{1}{2} = _$
1 blue = _____grays	$\frac{1}{4} = _$
1 pink = _____grays	$\frac{1}{6} = _$
1 brown = _____pinks	$\frac{1}{3} = _$
3 blues = _____grays	$\frac{3}{4} = _$